

# Cristian Mesquida

POSTDOCTORAL RESEARCHER

Eindhoven University of Technology, The Netherlands

+34 610 03 85 25 | [c.mesquida.caldentey@tue.nl](mailto:c.mesquida.caldentey@tue.nl) | [Cristian Mesquida](#) | [cristianmesquida](#)

## Employment

### Postdoctoral researcher

Eindhoven, The Netherlands

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025-Present

- Contributing to the development of Metacheck, an R package that helps researchers identify and prevent poor research practices before publication and facilitates data extraction from published articles

### PhD candidate

Eindhoven, The Netherlands

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2021-2025

- My PhD thesis examined the factors hindering replicability and reproducibility in sports and exercise science

## Education

### PhD

Eindhoven, The Netherlands

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2021-2025

- My PhD thesis focused on the factors that hinder replicability and reproducibility of research in sports and exercise science. You can find my thesis at: <https://research.tue.nl/en/publications/an-investigation-into-the-barriers-to-replicability-and-reproduci/>

### MSc. Integrative Physiology

Barcelona, Spain

UNIVERSITY OF BARCELONA

2017-2019

- My master thesis focused on the effect of two training protocols on maxymal oxygen uptake in highly-trained cyclists

### MSc. Research in Sport and Physical Activity

Granada, Spain

UNIVERSITY OF GRANADA

2015-2017

- My master thesis focused on the performance determinants of swimming sprint performance

### BSc. Sports and Exercise Science

Vic, Spain

UNIVERSITY OF VIC

2009-2013

- My bachelor end project focused on developing a training programme to reduce the injury risk of hamstrings in football players

## Publications

- Lakens, D., Cristian, Xavier-Quintais, G., Rasti, S., Toffalini, E., & Altoè, G. (2025). *Rethinking Type S and M Errors*. OSF. [https://doi.org/10.31234/osf.io/2phzb\\_v1](https://doi.org/10.31234/osf.io/2phzb_v1)
- Mesquida, C., Murphy, J., Warne, J., & Lakens, D. (2025). *The prevalence, reporting practices, and methodological quality of a priori power analyses in sports and exercise science research*. SportRxiv. <https://doi.org/10.51224/SRXIV.575>
- Mesquida, C., Warne, J., & Lakens, D. (2025). *A scoping review of the transparency, reporting practices and methodological rigor of meta-analyses published in Sports Medicine*. SportRxiv. <https://doi.org/10.51224/SRXIV.576>
- Mesquida, C., Warne, J., & Lakens, D. (2025). *What is your hypothesis? On the importance of knowing your hypothesis before conducting a hypothesis test*. SportRxiv. <https://doi.org/10.51224/SRXIV.577>
- Ditroilo, M., Mesquida, C., Abt, G., & Lakens, D. (2025). *Exploratory research in sport and exercise science: Perceptions, challenges, and recommendations*. *Journal of Sports Sciences*, 1–13. <https://doi.org/10.1080/02640414.2025.2486871>
- Mesquida, C., Murphy, J., Warne, J., & Lakens, D. (2025). *On the replicability of sports and exercise science research: Assessing the prevalence of publication bias and studies with underpowered designs by a z-curve analysis*. SportRxiv. <https://doi.org/10.51224/SRXIV.534>
- Murphy, J., Caldwell, A. R., Mesquida, C., Ladell, A. J. M., Encarnación-Martínez, A., Tual, A., Denys, A., Cameron, B., Van Hooren, B., Parr, B., DeLucia, B., Mason, B. R. J., Clark, B., Egan, B., Brown, C., Ade, C., Sforza, C., Taber, C. B., Kirk, C., ... Warne, J. P. (2025). *Estimating the Replicability of Sports and Exercise Science Research*. *Sports Medicine*. <https://doi.org/10.1007/s40279-025-02201-w>

8. Lakens, D., Mesquida, C., Rasti, S., & Ditroilo, M. (2024). The benefits of preregistration and Registered Reports. *Evidence-Based Toxicology*, 2(1), 2376046. <https://doi.org/10.1080/2833373X.2024.2376046>
9. Mesquida, C., & Lakens, D. (2024). Is the effect large enough to matter? Why exercise physiologists should interpret effect sizes meaningfully: A reply to Williams et al. (2023). *The Journal of Physiology*, 602(1), 241–242. <https://doi.org/10.1113/JP285901>
10. Holgado, D., Mesquida, C., & Román-Caballero, R. (2023). Assessing the Evidential Value of Mental Fatigue and Exercise Research. *Sports Medicine*, 53(12), 2293–2307. <https://doi.org/10.1007/s40279-023-01926-w>
11. Mesquida, C., Murphy, J., Lakens, D., & Warne, J. (2023). Publication bias, statistical power and reporting practices in the Journal of Sports Sciences: Potential barriers to replicability. *Journal of Sports Sciences*, 41(16), 1507–1517. <https://doi.org/10.1080/02640414.2023.2269357>
12. Murphy, J., Mesquida, C., & Warne, J. (2023). A Survey on the Attitudes Towards and Perception of Reproducibility and Replicability in Sports and Exercise Science. *Communications in Kinesiology*, 1(5). <https://doi.org/10.51224/cik.2023.53>
13. Bossi, A. H., Mesquida, C., Hopker, J., & Rønnestad, B. R. (2023). Adding Intermittent Vibration to Varied-intensity Work Intervals: No Extra Benefit. *International Journal of Sports Medicine*, 44(2), 126–132. <https://doi.org/10.1055/a-1812-7600>
14. Mesquida, C., Murphy, J., Lakens, D., & Warne, J. (2022). Replication concerns in sports and exercise science: A narrative review of selected methodological issues in the field. *Royal Society Open Science*, 9(12), 220946. <https://doi.org/10.1098/rsos.220946>
15. Murphy, J., Mesquida, C., Caldwell, A. R., Earp, B. D., & Warne, J. P. (2022). Proposal of a Selection Protocol for Replication of Studies in Sports and Exercise Science. *Sports Medicine*. <https://doi.org/10.1007/s40279-022-01749-1>
16. Etxebarría, N., Wright, J., Jeacocke, H., Mesquida, C., & Pyne, D. B. (2021). Running Your Best Triathlon Race. *International Journal of Sports Physiology and Performance*, 16(5), 744–747. <https://doi.org/10.1123/ijsp.2020-0838>
17. Bossi, A. H., Mesquida, C., Passfield, L., Rønnestad, B. R., & Hopker, J. G. (2020). Optimizing Interval Training Through Power-Output Variation Within the Work Intervals. *International Journal of Sports Physiology and Performance*, 15(7), 982–989. <https://doi.org/10.1123/ijsp.2019-0260>

## Teaching Experience

---

### Lecturer of Behaveoral Research Methods 3

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025-Present

- Teaching philosophy of science, power analysis, and equivalence testing. Teaching materials can be found at [https://cristianmesquida.github.io/multiple\\_regression/](https://cristianmesquida.github.io/multiple_regression/)

### Lecturer of Behaveoral Research Methods 2

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025-Present

- Teaching multiple regression and power analysis. Teaching materials can be found at [https://cristianmesquida.github.io/multiple\\_regression/](https://cristianmesquida.github.io/multiple_regression/) and

### Lecturer of Metascience

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024-Present

- Teaching Introduction to Meta-analysis as part of the Metascience course. Teaching materials can be found at [https://github.com/cristianmesquida/introduction\\_to\\_meta-analysis](https://github.com/cristianmesquida/introduction_to_meta-analysis)

### Teaching assistant of Behaveoral Research Methods 3

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024

- Assisting students during tutorial sessions with conducting their assignments

### Teaching assistant of Advanced Data Analysis

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024

- Taught R Markdown for reproducible reporting and data simulation for statistical analysis

### Teaching Assistant of Applied Exercise Physiology

TECHNOLOGICAL UNIVERSITY OF DUBLIN

2021

- Taught anatomy

# Supervision

---

## Supervisor of Master End Project

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025

- Co-supervised a project using automated search extraction to extract and attempted hypotheses using large language models

## Supervisor of Bachelor End Project

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025

- Co-supervised a meta-analysis project examining the impact of school sanitation on students' academic performance

## Supervisor of Master End Project

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024

- Co-supervised the development of a tool to identify incorrect use of the standard error of measurement instead of the standard deviation in effect size calculations for meta-analyses

## Supervisor of Bachelor End Project

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2023

- Co-supervised four students creating a tool to extract sample sizes from published studies

# Research experience

---

## Research internship (unpaid)

Norway

NORWEGIAN SCHOOL OF SPORTS SCIENCE

2018

- Together with Pr. Bjørn Harald, I investigated the use of the 1080 Sprint resistance device in swimming to measure power-velocity profiles and estimate drag characteristics in swimmers.

## Research internship (unpaid)

Norway

INLAND NORWAY UNIVERSITY OF APPLIED SCIENCES

2017

- Together with my supervisor Pr. Bent Rønnestad and my colleague Arthur Bossi, I investigated the effects of two high-intensity interval training (HIIT) protocols on maximal oxygen uptake ( $\text{VO}_2\text{max}$ ) in highly trained cyclists. The study involved detailed physiological assessments, including lactate threshold testing,  $\text{VO}_2\text{max}$  measurements, and cycling economy evaluations.

## Research internship (unpaid)

Spain

UNIVERSITY OF GRANADA

2015-2016

- Together with my supervisor Pr. Raúl Arellano, I contributed to race analysis and the evaluation of swimmers' dry-land performance profiles. Additionally, I assessed propulsive force using the 30-second tethered swimming test with a load cell force measurement system.

# Talks

---

## ONH Metascience Symposium

OSLO NYE HØYSKOLE

2026

- Talk slides: [https://cristianmesquida.github.io/metascience\\_symposium\\_oslo/#/title-slide](https://cristianmesquida.github.io/metascience_symposium_oslo/#/title-slide)

## Transparency, Technology and AI in Peer Review

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2026

- Talk slides: [https://cristianmesquida.github.io/technology\\_tue\\_conference\\_talk/#/title-slide](https://cristianmesquida.github.io/technology_tue_conference_talk/#/title-slide)

## Metascience Series

CHARLES STURT UNIVERSITY

2025

- Talk slides: <https://osf.io/bzp4s/files>

## Workshop on Metacheck

OPEN RESEARCH SUMMER SCHOOL AT KING'S COLLEGE LONDON

2025

- Workshop Materials: <https://osf.io/r8fbd/>

## Danish Reproducibility Network meetup

DANISH REPRODUCIBILITY NETWORK

2025

- Talk slides: <https://osf.io/tveh8/>

## Lunch meetup for the Open Science Community

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025

- Talk slides: <https://osf.io/tveh8/>

## NLRN Pre-Symposium for Young Meta-Scientists

UNIVERSITY MEDICAL CENTER GRONINGEN

2024

- Presented the findings from one of my PhD projects where we examined publication bias and average statistical power in sports and exercise science research

## Paul Meehl Graduate School PhD Day Conference

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024

- Presented the findings from one of my PhD projects where we examined the reporting practices and reproducibility of a priori power analyses in sports and exercise science research

## Workshop on endurance performance

SETANTA COLLEGE

2021-2023

- Conducted a workshop on the physiological determinants of endurance performance and team sports and conducted a lactate-threshold test to estimate training zones based on blood lactate concentrations

## Workshop on 1080 Sprint resistance device

KUMMING HIAGENG PHYSICAL TRAINING BASE

2019

- Conducted a workshop on the use of the 1080 Sprint resistance device to measure power-velocity profiles and estimate drag in swimmers.

# Grants

---

## Replication studies

DUTCH RESEARCH COUNCIL

2026

- Assess the reproducibility of power analyses in large-scale replication projects. See project details at <https://doi.org/10.61686/FSWVF35053>

## IE&IS ESA Education Innovation Grant

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024

- Funding to develop an R package designed to assist master's students and supervisors in performing automatic checks to prevent statistical errors and enforce best research practices